PHASE 1 KNOWLEDGE DOSSIER • WRITTEN OCTOBER 2026

Green Hydrogen, Without the Hype

What the molecule does for us every day, why the way we make it today has to end, and which machine I would trust to make it tomorrow.

WRITTEN BY

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Every number in here carries a bracketed source you can click, and the claims that matter carry a confidence tag: HIGH *I would stake the project on it,* MEDIUM *directionally right but sources disagree or are secondary,* LOW *treat as a working assumption.*

VISION

Clean molecules that pay their own way

Why I am doing this, and the three beliefs that shape every page that follows.

India already runs on hydrogen. It is in the urea that feeds the wheat, in the diesel that has had its sulphur stripped out, in the vanaspati in a dhaba kitchen. Almost none of it is clean. We make it by breaking natural gas, a large share of which we import, and we vent the carbon into the air as we do it. So we pay twice: once in dollars for the gas, once in the climate bill.

My vision for this project is short to say and hard to do: **make India's hydrogen clean where it matters most, and make it pay for itself.** Not clean as a slogan. Clean as a plant that a lender would fund, a fertiliser company would buy from, and a farmer would never notice except that the bag got greener.

Three beliefs sit under everything in this dossier.

1. **Green is not a business model. Value is.** If green hydrogen cannot beat grey on cost, or win a contract that pays for the difference, it is a press release, not a project. I judge every technology by the rupees of value it creates per rupee spent.
2. **The first customer is the fertiliser plant, not the car.** The hydrogen train is a fine symbol. The real prize is swapping the six-odd million tonnes of dirty hydrogen India already uses, starting with ammonia.
3. **Bet on technology the way an investor would.** Build the core on what is proven and bankable. Buy cheap options on what could change the game. Never confuse the two.

This is the first of four things Shalvi and I are building. It feeds a project-management presentation for Prof. Gopal, where we act as risk managers wearing project managers' badges; a research paper; a set of Python models on GitHub; and, much later, a blueprint for building a company from a single plant. This dossier does one job: it gives us the knowledge to pick the right machine. After it, we choose one stack and build everything else on that choice.

> *"Hydrogen is not the answer to everything. It is the only answer to a few things. My job is to find which few, and to price them honestly."*

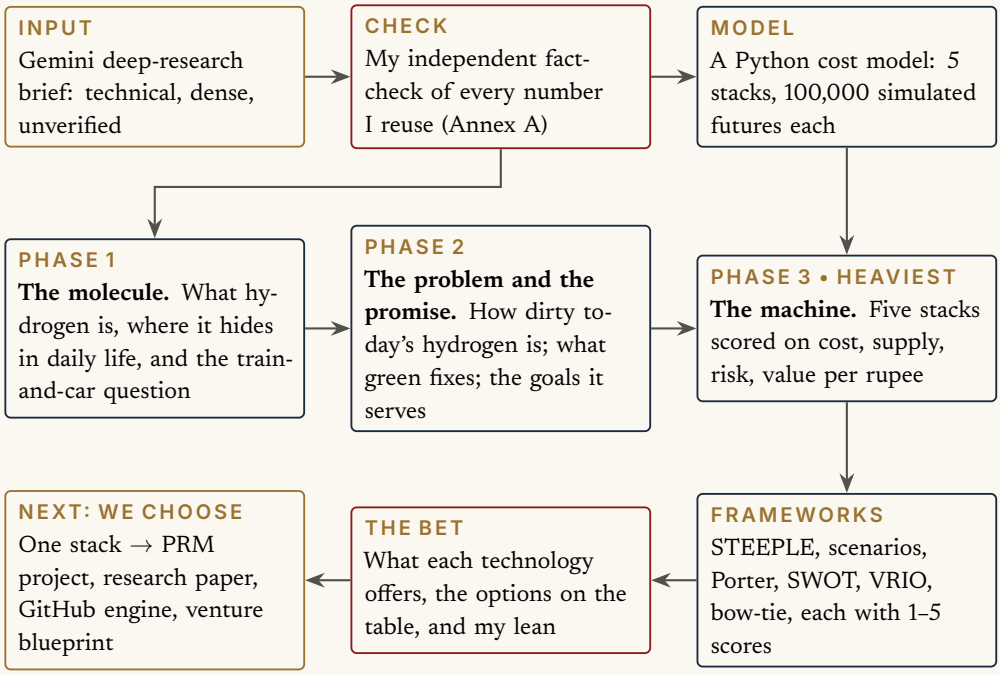
THREE NUMBERS TO KEEP IN YOUR HEAD

US$2.1–2.2 per kg is what grey hydrogen costs in India today [?, ?]. **About US$3.8 per kg** is what my model says a well-run Indian alkaline plant makes green hydrogen for. **About US$2.5–2.9 per kg** is what the winning green-ammonia bids in India's 2025 auctions quietly assume. The gap between the second and third numbers is the most important finding in this dossier, and Section **??** explains it.

WORK FLOW

The map of this dossier

Where the knowledge comes from, how it is checked, and where it goes next.



How I score things

Every framework in Phase 3 uses the same five-step scale, so that a "4" in Porter's model means the same strength as a "4" in the technology scorecard.

SCORE	WORD	WHAT IT MEANS FOR THE DECISION
●●●●● 5	Very high	Dominant. It would change the decision on its own.
●●●●○ 4	High	Strong. It shifts the decision unless something else outweighs it.
●●●○○ 3	Standard	Present, but it does not separate one option from another.
●●○○○ 2	Low	Weak. Worth noting, rarely decisive.
●○○○○ 1	Very low	Negligible, or a clear weakness when the scale is "higher is better".

In the technology scorecard every column is turned so that **5 is good for us.** A "5" under risk therefore means *low* risk. In Porter's model and the bow-tie, the score measures the *strength of the force or threat*, so a 5 there is bad news. I say which convention applies above each table.

Hydrogen, from the ground up

What it is, where it already hides in an ordinary Indian day, and the truth about hydrogen trains and cars.

1.1 The lightest thing there is, and why that is awkward

Hydrogen is the simplest atom in the universe: one proton, one electron. On Earth it almost never sits around on its own. It is locked inside water (H_2O) and inside fossil fuels such as natural gas (CH_4). So the first thing to understand is the one most headlines skip: **hydrogen is not an energy source. It is an energy carrier.** Someone has to spend energy to pull it out of water or gas, and the whole argument about "green" versus "grey" is really an argument about *where that energy comes from*.

The second thing is its odd personality. By weight it is a champion: a kilogram of hydrogen carries about 120 megajoules, nearly three times the 44 MJ in a kilogram of petrol. By volume it is a featherweight: even chilled into a liquid, a litre holds about 8 MJ against petrol's 32 MJ. To make it fit anywhere useful you either squeeze it to 350–700 times atmospheric pressure or cool it to −252.8 °C [?]. HIGH



Light to carry, bulky to store. That one fact explains why hydrogen suits big, fixed industrial plants far better than small moving vehicles. Source: [?].

1.2 Where hydrogen already touches your day

Most people think hydrogen is a fuel of the future. It is mostly a chemical of the present. The world used almost 100 million tonnes of it in 2024 [?], and in India it shows up in four places you would not expect.

FOOD
48%
of humanity was fed by fertiliser nitrogen from hydrogen-based ammonia by 2008 [?]. About 70% of all ammonia goes into fertiliser [?].

FUEL
2.6 Mt
of hydrogen a year goes into Indian refineries, mainly to strip sulphur out of petrol and diesel [?].

FAT
Vanaspati
is vegetable oil hardened by adding hydrogen, the cheaper cousin of ghee [?].

MATERIALS
Methanol
made from hydrogen feeds paints, resins and plastics; chlor-alkali, glass and electronics plants use it too [?].

The scale is the surprise. India used roughly 5.6 million tonnes of hydrogen made at home in 2018–19, plus about 1.1 million tonnes hidden inside the ammonia and methanol it imported, for a total near 6.7 million tonnes [?]. Nearly all of it is grey. Analysts expect demand to roughly double to about 12 million tonnes

by 2030, with fertiliser and refining still taking the lion's share [?]. MEDIUM (the base figures were modelled from chemistry rather than measured, and they are now seven years old.)

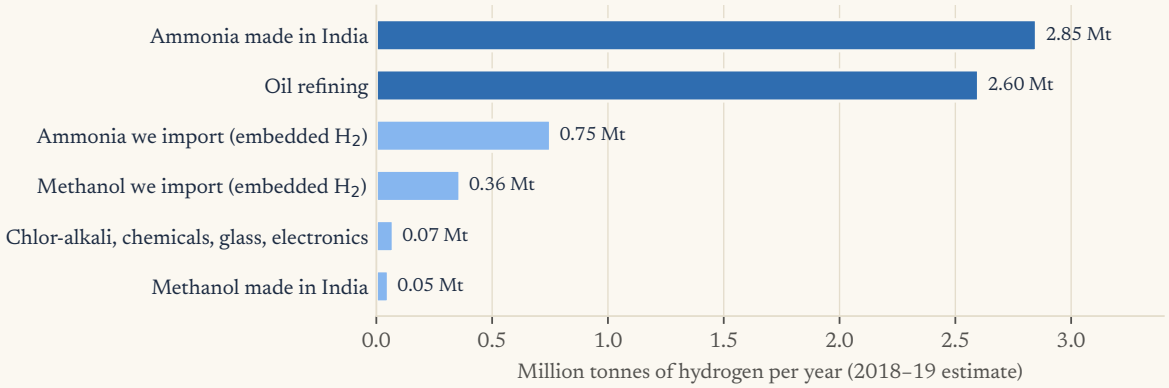


Figure 1. *India's hydrogen, by where it ends up. Two uses (fertiliser ammonia and refining) account for over 80%. Data: [?].*

IN PLAIN WORDS

If you ate a roti today, rode a bike on petrol, or bought biscuits made with vanaspati, you used hydrogen. You just never saw it, because it was made and consumed inside a factory fence. That is why the cleanest, quickest win is not a new hydrogen gadget but *changing how the hydrogen we already use gets made*.

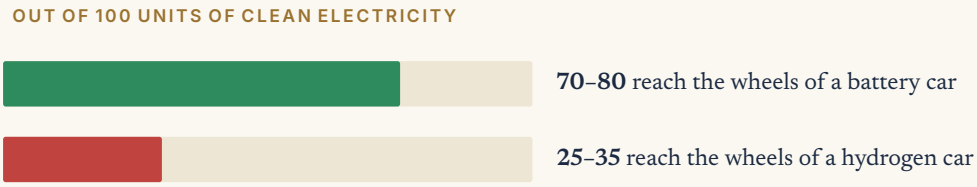
1.3 The hot topic: trains, cars and the hydrogen dream

What actually happened in Haryana. On 17 July 2026 the Prime Minister flagged off India's first hydrogen passenger train at Jind [?]. The facts are impressive. It is a 10-coach set carrying about 2,600 passengers over the 89 km between Jind and Sonipat. Two power cars of 1,200 kW each run on proton-exchange-membrane fuel cells, with a design speed of 110 km/h. The hydrogen is made on site at Jind by electrolysis, compressed to 500 bar, dispensed at 350 bar, and a store holds nearly 3,000 kg [?]. Officials call it the largest-capacity hydrogen train in service anywhere [?]. HIGH

What it proves, and what it does not. It proves Indian engineers can design, build and certify a hydrogen system end to end, including PESO safety approval [?]. That matters. What it does not prove is that hydrogen is the future of Indian rail. By November 2025, **99.2% of Indian Railways' broad-gauge network was already electrified** [?]. An electric train on an overhead wire uses the electricity directly; a hydrogen train turns electricity into hydrogen, then hydrogen back into electricity, and loses a lot on the way. So the hydrogen train's natural home is the last fraction of track: heritage and hill lines where wires are impractical, which is exactly where Indian Railways says it will deploy the technology [?]. HIGH

Germany ran this experiment first, and the lesson is sobering. Lower Saxony, home of the world's first hydrogen passenger trains, decided in 2023 that battery trains were cheaper and would replace the rest of its diesel fleet with them [?]. One analyst reports that by August 2025 only four of its fourteen hydrogen trainsets were running, for lack of replacement fuel-cell modules [?]. MEDIUM (opinion piece, but the 2023 battery decision is well documented.)

And hydrogen cars? Here the physics is unkind. Volkswagen's own engineers put it plainly: of the renewable electricity you start with, a battery car delivers about 70–80% to the wheels; a hydrogen fuel-cell car delivers about 25–35% [?]. The market has voted accordingly.



Source: [?]. Bars drawn at the range midpoints.

17 million+

electric cars sold worldwide in 2024 [?]

≈12,900

hydrogen fuel-cell vehicles sold in 2024 (16,011 in 2025) [?]

1 : 1,300

roughly one hydrogen car for every 1,300 electric cars [?]

The 2024 fuel-cell figure is back-calculated from SNE Research's 2025 total of 16,011 units and its 24.4% growth rate [?]; the ratio uses IEA's 17 million electric cars [?]. HIGH on direction, MEDIUM on the exact ratio.

Michael Liebreich's widely used "hydrogen ladder" reaches the same verdict from the other end. Its top row, labelled "no alternative", holds the jobs hydrogen already does, mostly fertiliser and refining. Its bottom row, which he calls "the row of doom", holds passenger cars, bulk e-fuels and power generation from hydrogen that was never stored [?]. HIGH

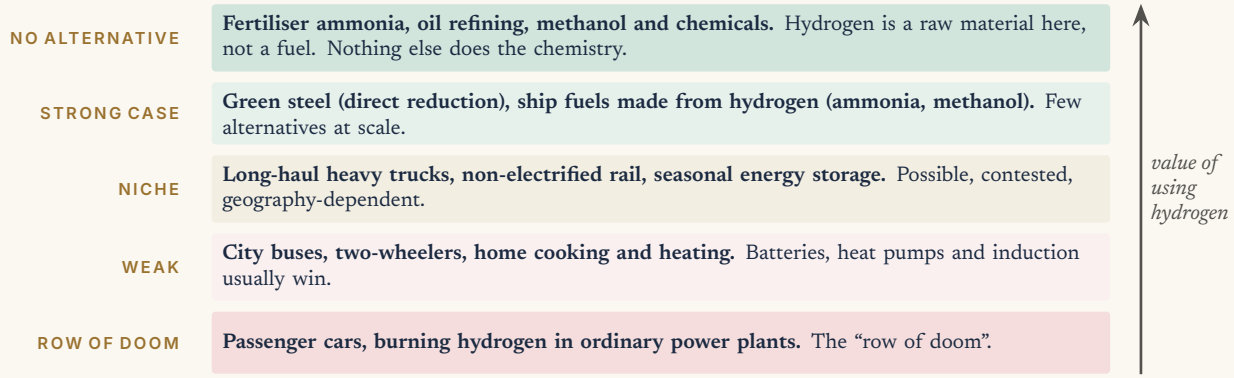


Figure 2. *My five-rung reading of where hydrogen earns its keep. Top and bottom rows match Liebreich's ladder v5.0 [?]; the middle rows are my own grouping.*

MY VERDICT ON THE DREAM

The vision of "hydrogen engines replacing petrol engines and EVs" is the wrong dream for India. For cars, scooters and city buses, batteries win on physics, and the market is already settled. The *right* dream is bigger and less visible: make the six-odd million tonnes of hydrogen India already uses clean, then use clean hydrogen for the jobs electricity cannot do (steel, ships, fertiliser). That is where it cuts emissions and imported gas at the same time. HIGH

1.4 Is our project connected to the train and the car?

Yes, more than it looks. The Jind refuelling station is a miniature of exactly what we would build: electrolysers, compressors and storage tanks next to a source of power. Our plant would be the same idea at roughly a thousand times the scale.

A quick sense of size, using my base-case model (Section 3): a 100 MW alkaline plant running about 5,000 hours a year at 53 kWh per kilogram would make about **9,400 tonnes of hydrogen a year, or 26 tonnes**

a day. Turned into ammonia, that is about **53,000 tonnes a year**, almost exactly the size of one mid-sized SECI green-ammonia contract [?]. The same 26 tonnes a day could refuel about 200 German-style hydrogen trains, each needing about 130 kg [?]. MEDIUM

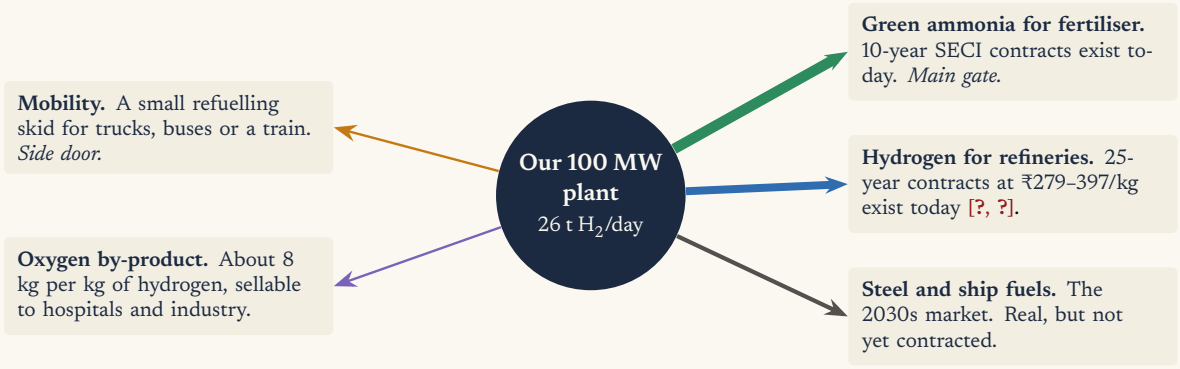


Figure 3. *Where our hydrogen could go. Arrow thickness shows my ranking of near-term value, not exact volumes. The oxygen ratio follows from the chemistry of splitting water (8 kg O_2 per kg H_2).*

SO CAN WE HELP FUEL THE HYDROGEN TRAIN OR A HYDROGEN BUS?

Technically, yes. A compressor and a dispenser bolted onto our plant would do it. Commercially, it should be a side door, not the main gate. A single train or bus fleet would use a fraction of one percent of our output, and its demand is uncertain. The customer who can sign a long contract for most of our output is a fertiliser company or a refinery. We build for them, and keep the mobility skid as a cheap option.

Today's hydrogen is dirty, and India pays for it twice

How hydrogen is made now, what that costs the climate and the exchequer, what "green" really promises, and the goals it serves.

2.1 How the world makes hydrogen today

The world used almost 100 million tonnes of hydrogen in 2024, and the vast majority came from fossil fuels with no carbon capture at all [?]. In 2023 nearly two-thirds was made from natural gas and about a fifth from coal [?]. HIGH

The usual method is *steam methane reforming*. In plain words: you heat natural gas with steam until the molecules break apart, keep the hydrogen, and release the carbon as CO_2. The International Energy Agency (IEA) puts the damage at **10–12 kg of CO_2-equivalent for every kilogram of hydrogen made from gas, and 22–26 kg when it is made from coal** [?].

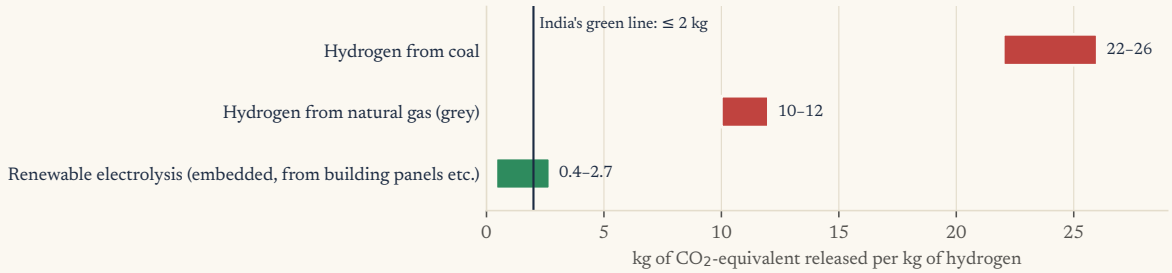


Figure 4. *Carbon released per kilogram of hydrogen, by production route. Data: [?]; India's threshold: [?].*

Add it up and hydrogen production released about **920 million tonnes of CO_2 in 2023** [?]. Against the 37.4 billion tonnes of energy-related CO_2 the world emitted that year [?], that is roughly 2.5%, more than most countries emit in total. Ammonia alone accounts for about 450 million tonnes of direct emissions a year, roughly 2.4 tonnes of CO_2 per tonne made, which the IEA notes equals the whole energy system of South Africa [?]. HIGH

2.2 The double bill India pays

BILL ONE • THE CARBON BILL

55–70 Mt

of CO_2 a year from the hydrogen India makes at home. My arithmetic: about 5.6 Mt of grey hydrogen [?] × 10–12 kg CO_2 per kg [?]. The Mission's own target of 50 Mt avoided for 5 Mt of green hydrogen implies the same ratio [?].

BILL TWO • THE DOLLAR BILL

₹1.86 lakh crore

fertiliser subsidy in FY26 (revised), with urea alone at ₹1.26 lakh crore [?]. Roughly half of India's gas is imported [?, ?], and the fertiliser sector takes 58% of the imported LNG India regasifies [?]. Urea imports may have topped 10 Mt in FY26 [?].

Confidence in both bills: MEDIUM *(the carbon figure is my arithmetic on 2018–19 volumes; the subsidy figure comes from a news digest of the Union Budget).*

Here is the part that took me a while to see. The farmer does not feel a gas-price shock, because the retail price of urea has been frozen at ₹5,360 a tonne since November 2012 [?]. **The government absorbs the shock instead.** When the rupee slid from about 86.6 to 91.9 to the dollar, the subsidy bill rose with it [?]. So the real owner of India's grey-hydrogen risk is the exchequer.

> KEY FINDING • GREEN HYDROGEN IS A HEDGE BEFORE IT IS A CLIMATE TOOL
>
> Grey hydrogen's cost moves with imported gas and the rupee. Green hydrogen's cost is mostly capital, fixed on the day the plant is financed. When SECI signs a 10-year fixed-price green-ammonia contract [?], the state is really buying an *insurance policy*: it swaps a volatile, dollar-linked cost for a known rupee number. For a finance student this is the cleanest way to explain why a "more expensive" molecule can still be the cheaper choice. HIGH on the mechanism, MEDIUM on its size.

2.3 What "green" actually promises

India wrote its definition down in August 2023. Hydrogen counts as **green** only if it is made from renewable energy, by electrolysis or by converting biomass, and emits **no more than 2 kg of CO_2-equivalent per kg of hydrogen**. That is measured "well-to-gate", covering water treatment, electrolysis, purification, drying and compression, and averaged over the previous 12 months. The Bureau of Energy Efficiency accredits the agencies that check it [?]. HIGH

Three honest caveats keep the promise from becoming a slogan.

- **Building the panels has a footprint.** Manufacturing the renewable plant embeds 0.4–2.7 kg CO_2 per kg of hydrogen, which most standards do not count [?]. HIGH
- **Leaks warm the planet.** A multi-model study put hydrogen's 100-year global warming potential at about 11.6 times that of CO_2 [?]. Hydrogen does not trap heat directly, but in the air it lengthens the life of methane and other warming gases. A leaky green plant is less green than it looks. MEDIUM
- **Water is a smaller problem than people fear, but location matters.** Green hydrogen uses about 20–30 litres per kg once purification and cooling are counted, on par with or below grey hydrogen's 20–40 litres [?]. The issue is not the total but *which district* the water comes from. Coastal sites with desalination avoid the fight. HIGH

Table 1. *The hydrogen "colours" that matter for an Indian decision.*

COLOUR	HOW IT IS MADE	CO_2 PER KG H_2	COST AND STATUS IN INDIA	CONF.
Grey	Natural gas split with steam, carbon vented	10–12 kg [?]	About US$2.1–2.2/kg [?, ?]. The default today.	HIGH
Black/brown	Coal gasified, carbon vented	22–26 kg [?]	Minor in India; the dirtiest route.	HIGH
Blue	Grey plus carbon capture and storage	Lower, depends on capture rate and methane leaks	Capture costs US$60–110 per tonne of CO_2 [?], adding roughly US$0.3–1.2/kg at 55–90% capture (my arithmetic). I found no large Indian CO_2 storage hub to send it to.	LOW
Green (electrolysis)	Water split with renewable power	≤2 kg by law [?]	US$3.5–4/kg today [?]; refinery contracts at ₹279–397/kg [?, ?].	HIGH
Green (biomass)	Crop waste or biogas converted to hydrogen	Counts as green under India's standard [?]	Early pilots; limited by collecting enough feedstock.	LOW

2.4 The goals it serves

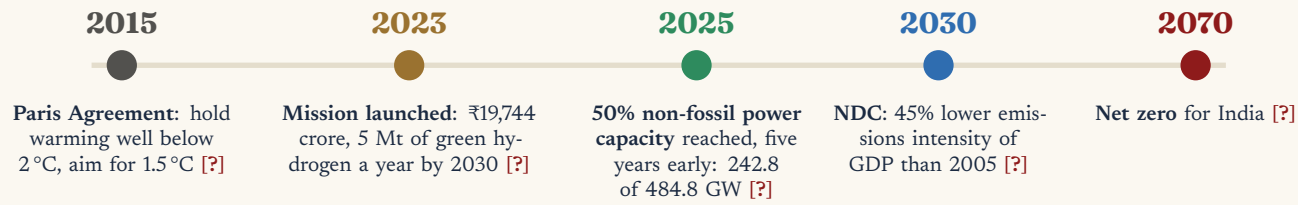


Figure 5. *The commitments green hydrogen is meant to serve.*

The National Green Hydrogen Mission is the bridge between these targets and our plant. Its promised 2030 outcomes are worth quoting exactly, because every one of them is a stakeholder interest we will meet again in the project: **at least 5 million tonnes a year of green hydrogen, about 125 GW of new renewable capacity, over ₹8 lakh crore of investment, over 6 lakh jobs, over ₹1 lakh crore of cumulative fossil-fuel import savings, and about 50 million tonnes a year of greenhouse gas avoided** [?]. Of the ₹19,744 crore outlay, ₹17,490 crore sits in SIGHT, the incentive scheme that funds electrolyser factories and green hydrogen and ammonia production [?]. HIGH

Two more goals give the case teeth. First, the UN Sustainable Development Goals: green hydrogen speaks directly to clean energy (SDG 7), industry and infrastructure (SDG 9) and climate action (SDG 13), and through fertiliser security to zero hunger (SDG 2) [?]. Second, Europe's border carbon charge. The definitive phase of the EU's Carbon Border Adjustment Mechanism began in January 2026 and covers both hydrogen and fertilisers, including ammonia [?]. Any Indian exporter of grey ammonia to Europe will eventually pay for its carbon; a green exporter will not. HIGH

2.5 Promise versus delivery

Ambition is cheap. Here is the scoreboard as of 2026.

862,000 t

a year of green hydrogen capacity allocated to 19 companies [?]

≈8,000 t

a year actually commissioned by February 2026 [?], under 1% of the allocation

≈9%

utilisation of the world's 58 GW a year of electrolyser factories in 2025 [?]

Globally, 2025 was a correction. Developers committed to roughly 1 million tonnes a year of new low-emission capacity but cancelled over 4.9 million tonnes' worth [?]. The IEA cut its estimate of 2030 low-emission output from 49 to 37 million tonnes a year [?]. Oman's US$6.7 billion HyDuqm project was dropped after its partners reviewed "offtake dynamics", which is polite language for "nobody signed to buy" [?]. HIGH

MY VERDICT ON ``FUTURE-PROOF AND NECESSARY''

Necessary: yes, for the fertiliser, refining, steel and shipping jobs that nothing else can do, and because it turns an imported, volatile cost into a domestic, fixed one. *Future-proof*: only conditionally. The projects that survived 2025 had one thing in common, a creditworthy buyer under a long contract. Green hydrogen is a future-proof idea; any single green hydrogen plant is only as future-proof as its offtake contract. That sentence becomes the first rule of our project. HIGH

Choosing the electrolyser stack

Five ways to split water, judged on market availability, cost, supply chain, risk and the value each rupee buys.

3.1 What a green hydrogen plant actually is

The electrolyser gets the headlines, but it is only the heart of a much larger body. A green-ammonia plant is really three plants bolted together: a power plant, a hydrogen plant and a chemical plant.

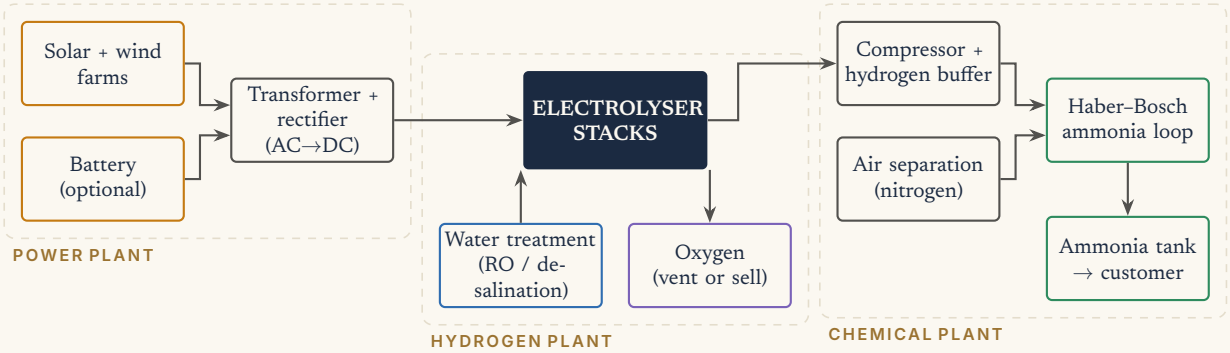


Figure 6. *The parts of a green-ammonia plant. Everything except the dark box is "balance of plant", and it is where cost overruns usually hide.*

IN PLAIN WORDS

The price tag on the electrolyser is not the price of the plant. The IEA finds that a Chinese-made electrolyser installed in China costs US$500–1,100 per kW all-in, while systems made outside China cost US$1,900–2,500 per kW installed. Once a Chinese unit is shipped abroad, engineering and contingency costs "narrow the gap sharply" [?]. So when someone quotes a US$300 stack, ask what the transformer, compressor, water plant, civil works and contingency add. HIGH

3.2 The five stacks, in plain words

There are four basic ways to split water with electricity, and one sensible blend of two of them. I treat each as a "stack" we could build the project on.

Table 2. *The five candidate stacks. Efficiency, life and materials are IRENA's 2020 state of the art [?]; flexibility from [?]; 2025–26 status from the sources cited.*

	HOW IT WORKS	EFFICIENCY & LIFE	FLEXIBILITY & MATERIALS	WHERE IT STANDS, 2026
Alkaline (pressurised)	Nickel-coated steel electrodes in a caustic potash bath, 70–90 °C	50–78 kWh/kg; stack about 60,000 h	Pressurised units turn down only to about 40% load, atmospheric to 15%; nickel and steel, no precious metals	The workhorse. Most Chinese makers build it [?]; a 1 MW unit fell from about US$800 to US$400/kW in three years [?]. HIGH
PEM	A thin acid polymer membrane; iridium oxide and platinum catalysts, 50–80 °C	50–83 kWh/kg; 50,000–80,000 h	Follows solar and wind closely; needs very pure water; relies on iridium	Commercial, pricier. Iridium is the long-run choke point (Section ??). HIGH
Hybrid ALK + PEM	Alkaline for the steady base, a PEM slice for the swings	Weighted mix of the two	Captures more renewable hours than alkaline alone	An engineering choice, not a product. Needs two OEMs and good controls. MEDIUM
AEM	A membrane like PEM, but alkaline chemistry with cheap nickel catalysts, 40–60 °C	57–69 kWh/kg; durability was above 5,000 h in 2020	Promises PEM-like flexibility without precious metals	First 5 MW system delivered in April 2026; an Indian 250–300 MW factory announced via a US licence [?]. Durability is the open question. LOW
SOEC	Ceramic cells at 700–850 °C splitting steam	45–55 kWh/kg (plus heat); under 20,000 h	Must run hot and steady; ceramics and perovskites	Leading maker Topsoe will "hibernate" its factory until demand is robust and dropped a planned US plant (June 2026) [?]. HIGH

3.3 Market availability: a buyer's market for proven kit

The electrolyser industry has built far more factories than the world is buying from. Global manufacturing capacity reached almost 58 GW a year by the end of 2025, about 60% of it in China, 20% in Europe and 10% in the US. Actual output was under 5 GW, a utilisation of roughly 9%, and the IEA says the sector is "entering a consolidation phase"; Cummins, for one, has stopped selling electrolysers once existing orders are filled [?]. China also hosts about 65% of the electrolyser capacity that is installed or at final investment decision worldwide [?]. India has awarded 3,000 MW a year of electrolyser manufacturing capacity to 15 firms under its incentive scheme [?]. HIGH

> **WHAT THE GLUT MEANS FOR US**
>
> For alkaline and PEM, we are the customer in a buyer's market. We can demand long warranties, performance guarantees and service agreements, because suppliers need the orders. The flip side is *supplier survival risk*: an OEM that goes bust in year six cannot honour a 10-year stack guarantee. For AEM and SOEC the market is the opposite: thin, early and unproven. MEDIUM

3.4 Cost of production: what a kilogram really costs

I built a small model to put all five stacks on the same footing. It asks one question: *over 25 years, what does each kilogram of hydrogen cost once every rupee of capital, maintenance, stack replacement and electricity is*

counted? Economists call this the levelised cost of hydrogen (LCOH):

$$\text{LCOH} = \frac{\overbrace{\text{Capex} \times \text{CRF}}^{\text{paying back the plant}} + \overbrace{\text{O\&M}}^{\text{upkeep}} + \overbrace{\text{Stack replacement}}^{\text{wear and tear}}}{\text{kg of hydrogen made per year}} + \underbrace{\text{Electricity price} \times \text{kWh per kg}}_{\text{the power bill}} + \text{water}$$

The capital recovery factor (CRF) turns a one-time cost into an equal yearly payment at the cost of capital, exactly like an EMI on a home loan. The full input table is in Annex B. The four inputs that matter most are:

- **Electricity:** US$32–48 per MWh, base US$38. The low end is the ₹2.86/kWh discovered for solar plus four-hour storage in October 2025 [?]. Round-the-clock firm power costs ₹5.25/kWh (about US$58) [?], which I use only for SOEC because it cannot follow the sun. MEDIUM
- **Cost of capital:** 8.5–12.5%, base 10%. This is my assumption; Gemini's brief estimated 9.5–11.5% [?]. LOW
- **Renewable hours:** an oversized solar-plus-wind farm can feed the stack 55–68% of the hours in a year. Alkaline captures 92% of those hours because of its high minimum load; PEM captures 99%. LOW
- **Installed cost:** alkaline US$800–1,600 per kW, PEM US$1,300–2,400, AEM US$1,000–3,000, SOEC US$2,800–5,600, all anchored on the IEA and IRENA ranges [?, ?, ?]. MEDIUM

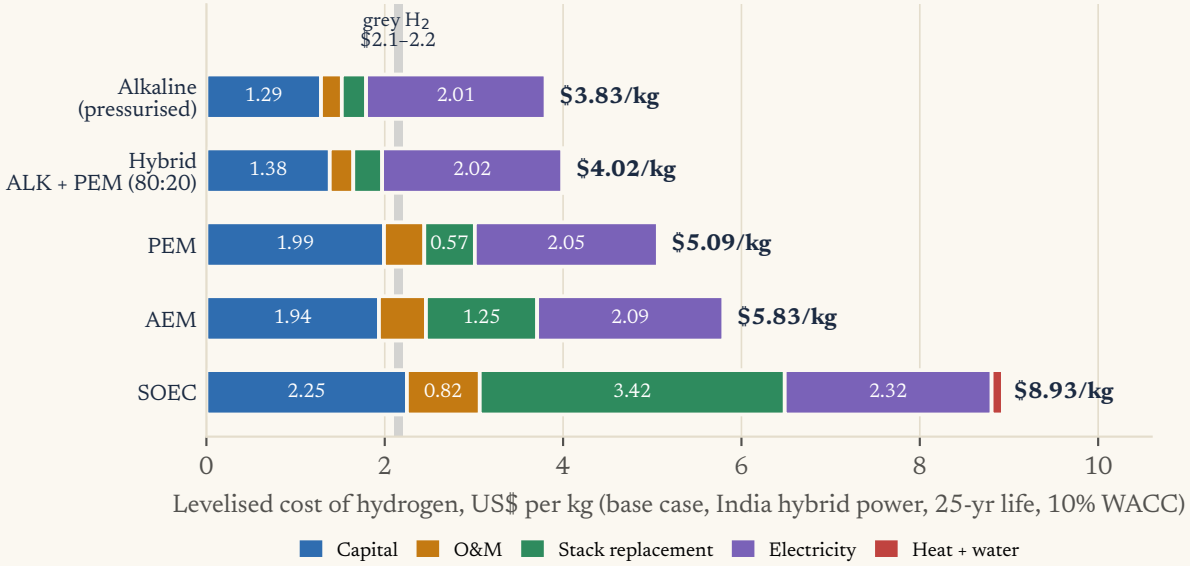


Figure 7. *Base-case cost of a kilogram of green hydrogen, split by where the money goes. Grey hydrogen in India: [?, ?].*

The base case puts **alkaline at US$3.83/kg**, the hybrid US$0.19 dearer, PEM US$1.26 dearer, AEM US$2.00 dearer and SOEC more than double. That alkaline figure sits inside the US$3.5–4/kg that market analysts quote for Indian green hydrogen today [?], which is my main check that the model is calibrated. MEDIUM

A single number hides the risk, so I ran each stack through 100,000 simulated futures, drawing every uncertain input from its low–likely–high range (Figure ??).

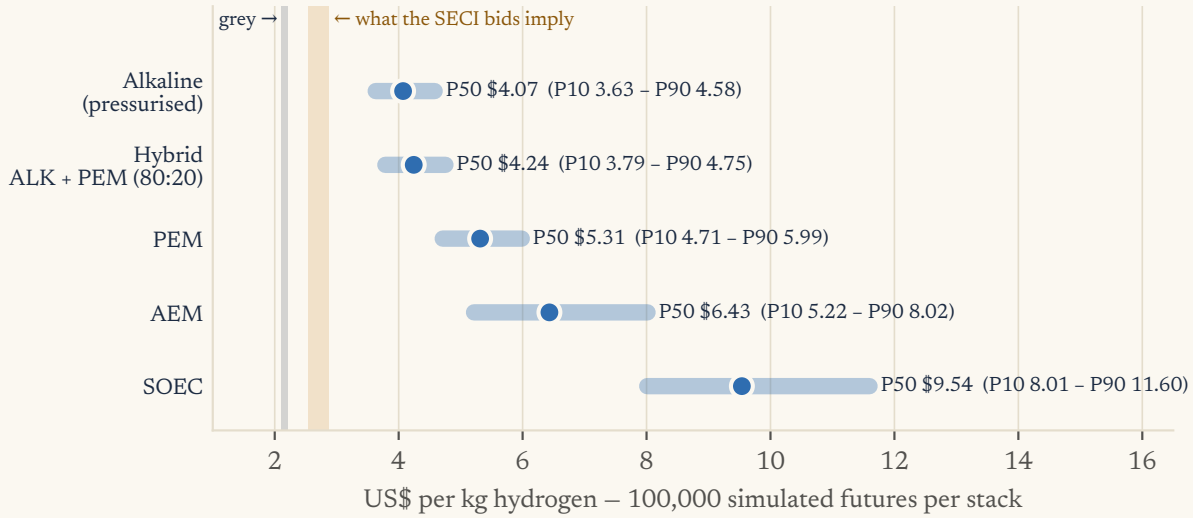


Figure 8. *Monte Carlo results: the bar spans the 10th to 90th percentile, the dot is the median (P50).*

Two things jump out. First, **no stack gets below US$3/kg in any meaningful share of futures**; under my ranges the probability is effectively zero for all five. Second, every median sits *above* its base case (alkaline US$4.07 against US$3.83). That is not a bug. Cost overruns are more likely than underruns, so the ranges are skewed to the right, and the "most likely" input does not give the "middle" outcome. Any project manager who budgets on the most-likely value will overspend more often than not. HIGH

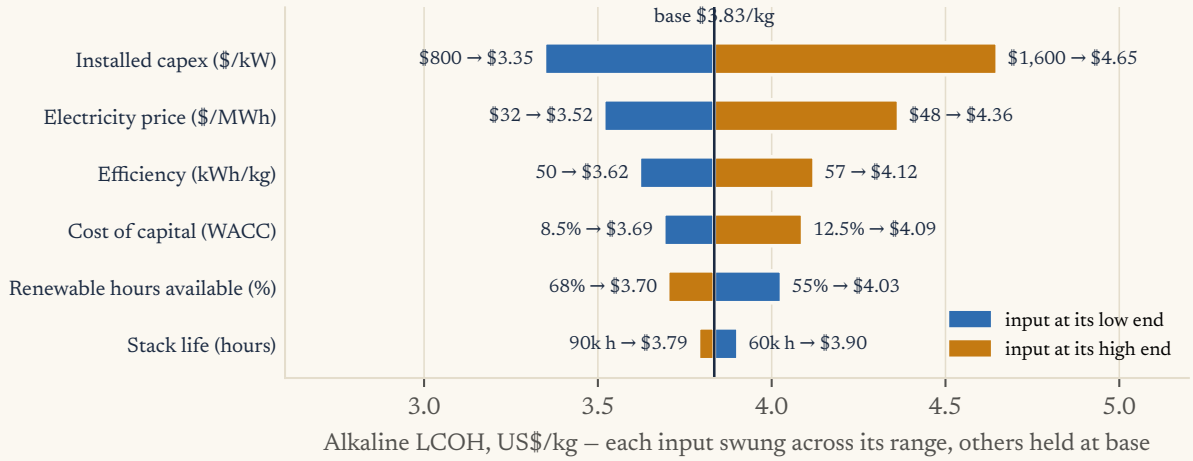


Figure 9. *Which input moves the alkaline cost most. Each bar swings one input across its full range, holding the rest at base.*

> **THE BIGGEST LEVER IS A PROJECT-MANAGEMENT LEVER**
>
> Installed capital cost swings alkaline hydrogen by US$1.30/kg, more than the electricity price (US$0.84) and far more than stack life (US$0.11). In other words, **the cost of green hydrogen is decided less in the lab than on the construction site**: in the engineering contract, the estimate's maturity and the control of scope creep. That is exactly the territory of our PRM project. HIGH

What the market is betting. India has real prices to test the model against. The thirteen green-ammonia auctions under SIGHT Mode 2A, Tranche I, allocated all 724,000 tonnes a year on 10-year fixed-price contracts [?].

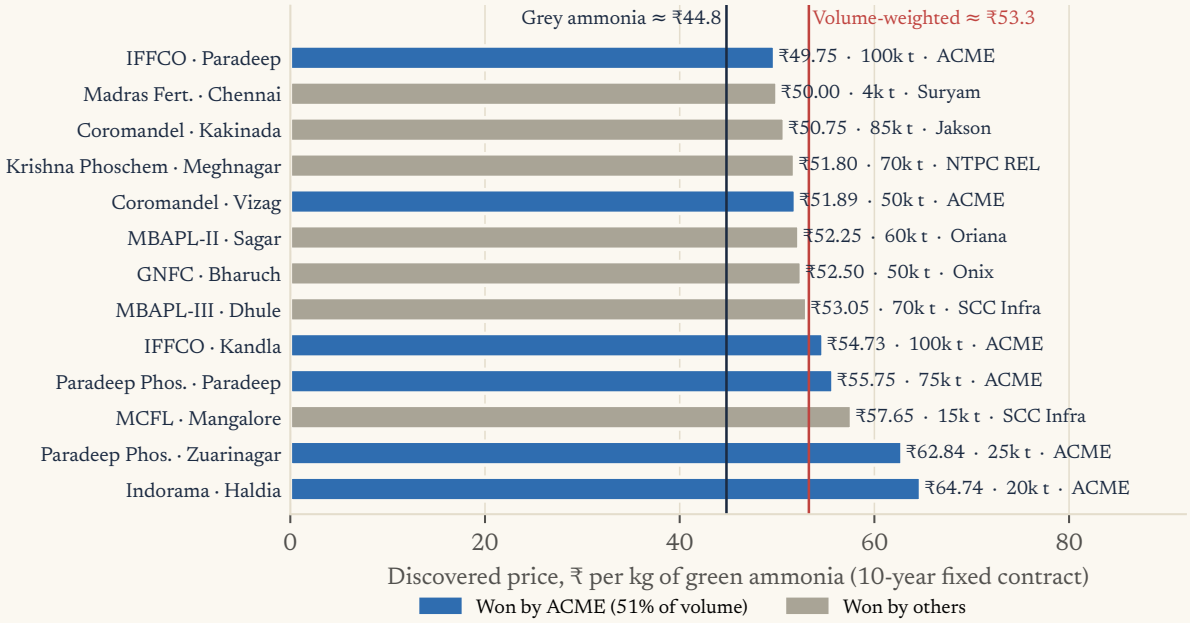


Figure 10. *All 13 SECI green-ammonia auctions (2025), winners and discovered prices. Data: SECI result sheet [?]; grey ammonia benchmark US$515/t [?] converted at ₹87.*

The volume-weighted price is **₹53.27 per kg of ammonia, about US$612 a tonne**, roughly 19% above grey ammonia's US$515. (JMK Research, using local prices, puts the lowest bid only 10% above grey [?].) ACME alone won 51% of the volume. Bids were razor-close: at IFFCO Kandla the top three were ₹54.73, ₹54.98 and ₹55.48 [?]. A modest top-up helps: ₹8.82, ₹7.06 and ₹5.30 per kg of ammonia in years one to three [?]. HIGH

Strip out a reasonable US$100–160 per tonne for making ammonia from hydrogen (my assumption, LOW), divide by the 0.178 tonnes of hydrogen in each tonne of ammonia, and the bids imply hydrogen at roughly **US$2.5–2.9/kg**. That is a full dollar below my base case. So I asked the finance question: *what would you have to believe for that price to make money?*

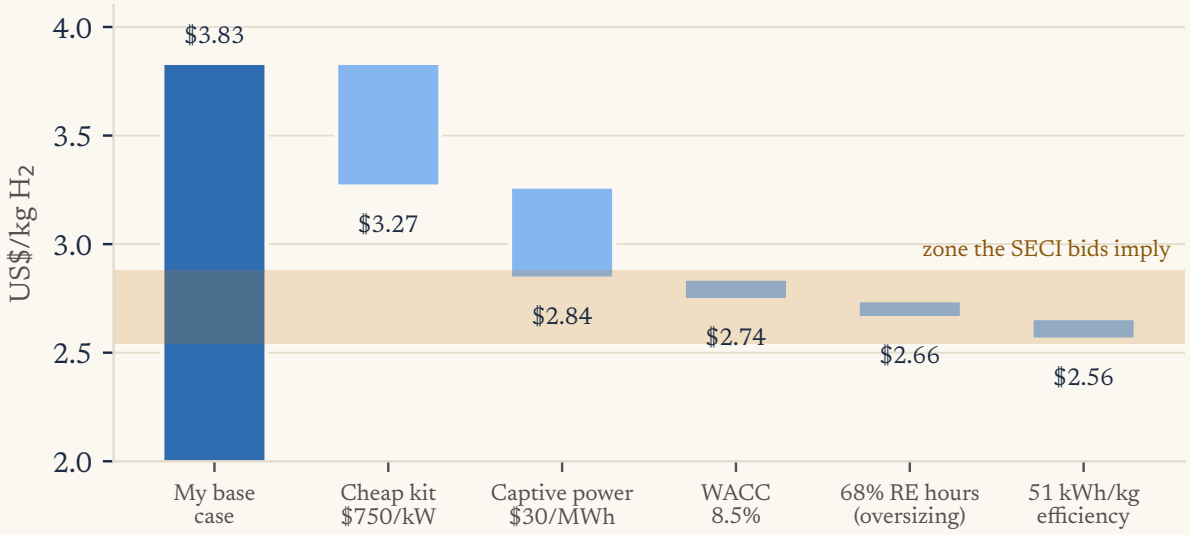


Figure 11. *"What you have to believe": stacking favourable assumptions on the alkaline base case until it reaches the price the auctions imply.*

KEY FINDING • THE WINNING BIDS NEED ALMOST EVERYTHING TO GO RIGHT

To reach the auction-implied cost, a developer needs cheap equipment (about US$750/kW installed), captive power near US$30/MWh, a cost of capital near 8.5%, heavy oversizing to run 68% of hours, *and* best-in-class efficiency, all at once. Large groups with their own renewable fleets and balance sheets may get there. For a new entrant it leaves almost no buffer. In auction theory this is the **winner's curse**: in a tight reverse auction, the winner is often the bidder who was most optimistic, not the one with the lowest true cost. Delivery risk on these contracts deserves close watching. MEDIUM

The refinery market tells a similar story with longer contracts. Indian Oil's Panipat tender went to L&T at ₹397/kg (about US$4.6) for 25 years [?]; Numaligarh Refinery's went to NeuEN Green Energy at ₹279/kg (about US$3.1) in February 2026 [?]. HIGH

3.5 Supply-chain feasibility

THE IRIDIUM ARITHMETIC

PEM electrolysers need iridium, one of the rarest metals mined. The world produces only about 7–8 tonnes a year, around 95% of it from South Africa and Zimbabwe, and today's PEM designs use roughly 400 kg per gigawatt [?]. Do the division: **even if every gram of the world's iridium went into PEM, it would cover only about 17–20 GW a year**. At a realistic 20–40% share, that is roughly 3.5–8 GW a year (my arithmetic). Cutting iridium use by 75% per gigawatt, which industry believes is achievable, would quadruple that [?]. MEDIUM

Table 3. *Supply-chain exposure by stack. Score: 5 = easy to source, diversified; 1 = fragile.*

STACK	CRITICAL INPUT AND WHERE IT COMES FROM	INDIAN POSITION	SCORE
Alkaline	Nickel, steel, diaphragms; makers concentrated in China (about 60% of global factory capacity) [?]	Domestic factories under the 3 GW incentive scheme [?]	●●●●○
Hybrid	Alkaline inputs plus a small PEM slice	As alkaline, plus a modest import	●●●●○
PEM	Iridium and platinum, roughly 95% from southern Africa [?]	Import-dependent for catalysts	●●○○○
AEM	Nickel and iron catalysts, no precious metals [?]; supplier base tiny	One licensed factory announced [?]	●●●○○
SOEC	Ceramics, perovskites; very few makers	None	●●○○○

Three cross-cutting supply risks apply to every stack. **Currency**: imported kit is priced in dollars, and the rupee moved from about 86.6 to 91.9 per dollar over 2025 [?]. **Long-lead items**: Gemini's brief puts large compressors at 18–24 months and switchgear at 12–16 months [?]; I could not verify this LOW. **Water**: desalination adds at most about US$0.10/kg and about 1% of the electricity [?], so a coastal site turns water from a political risk into a small cost. HIGH

3.6 Technical and operational risk

The core operational problem is simple to state: **the sun and wind are variable, and the ammonia loop wants a steady diet**. Each stack copes differently.

Table 4. *Technical and operational risk. Score: 5 = low risk for us; 1 = high risk.*

STACK	MAIN TECHNICAL RISK	HOW WE WOULD MANAGE IT	SCORE
Alkaline	Slow ramping; a pressurised unit idles below about 40% load and wastes the morning and evening shoulders of solar [?]	Oversize renewables, add a battery or hydrogen buffer; atmospheric units can turn down to 15%	●●●●○
Hybrid	Two technologies, two OEMs, one control system	One integrator under a single performance guarantee	●●●○○
PEM	Shorter stack life; ultra-pure water; iridium cost at stack replacement	Long-term service agreement with degradation guarantees	●●●○○
AEM	Durability not yet proven at multi-MW scale in the field [?]	Pilot-scale only until field hours exist	●○○○○
SOEC	Under 20,000 h stack life, needs constant high heat [?]; maker pausing production [?]	Only beside an industrial heat source	●○○○○

Safety applies to all five. Hydrogen leaks easily and embrittles some steels, and ammonia is toxic. Indian plants need approval from the Petroleum and Explosives Safety Organisation (PESO), as the Jind station did, along with leak detection, ventilation and automatic shut-off [?]. HIGH

3.7 Value created per rupee spent

To make "value per rupee" concrete I use one plain question: *for every ₹100 we spend making green hydrogen, how many rupees' worth of grey hydrogen, and of the carbon it would have released, do we replace?* If the answer were above ₹100, green hydrogen would pay for itself with no contract or subsidy.

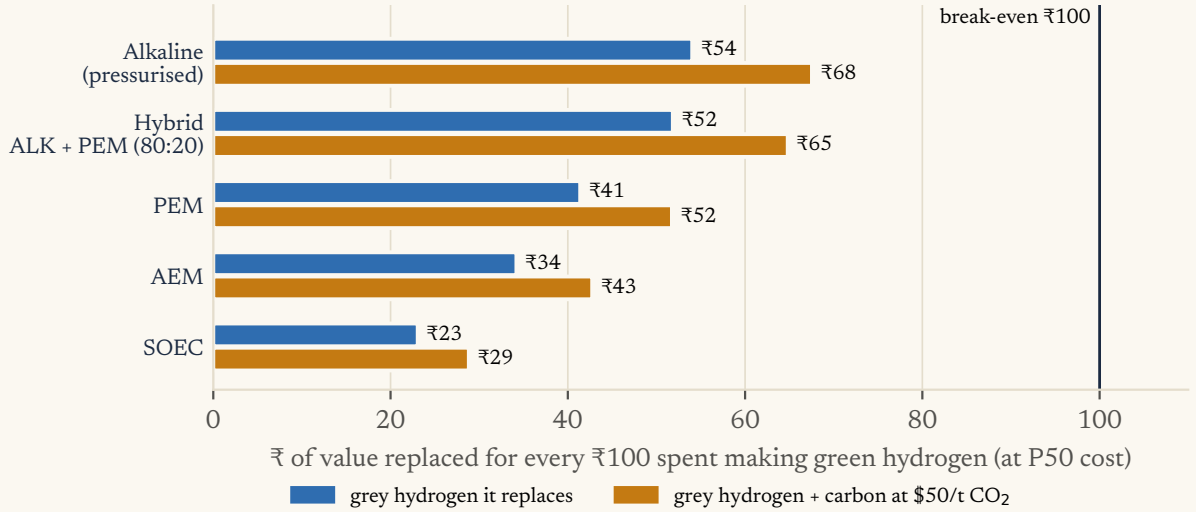


Figure 12. *Value replaced per ₹100 spent, at each stack's P50 cost. Grey hydrogen at US$2.2/kg [?]; 11 kg CO_2 per kg [?]; the US$50/t carbon price is illustrative, not an Indian market price.*

Table 5. *What ₹1 crore buys over the plant's life, at P50 cost and ₹90/US$.*

STACK	HYDROGEN PER ₹1 CRORE	CO_2 AVOIDED PER ₹1 CRORE	VALUE PER ₹100 (WITH CARBON)
Alkaline (pressurised)	27.3 t	300 t	₹54 (₹68)
Hybrid ALK + PEM	26.2 t	288 t	₹52 (₹65)
PEM	20.9 t	230 t	₹41 (₹52)
AEM	17.3 t	190 t	₹34 (₹43)
SOEC	11.6 t	128 t	₹23 (₹29)

KEY FINDING • NO STACK PAYS FOR ITSELF YET, SO THE CONTRACT IS THE BUSINESS

Even the best stack replaces only about ₹54 of grey hydrogen for every ₹100 spent, or ₹68 if carbon is priced at US$50 a tonne. The gap has to be closed by a contract (SECI's fixed price, a refinery's 25-year deal), a subsidy, or a carbon price. This is why I keep saying the offtake contract *is* the business. The technology choice decides how big a gap the contract has to close. HIGH

3.8 The scorecard

Figure **??** pulls the five criteria together and adds two more: how well a stack fits India's solar-and-wind supply, and its future upside. Every column is turned so that **5 is good for us**.

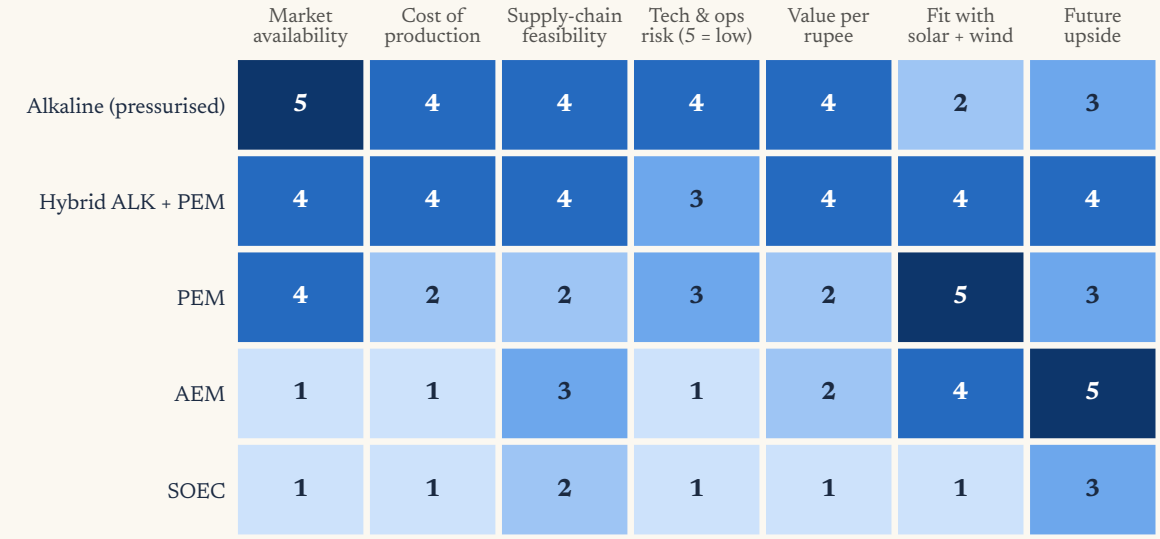


Figure 13. *Technology scorecard (1 = very low, 5 = very high; higher is better in every column). Each score is justified in Sections 3.2–3.7.*

How you weight the columns depends on who you are. A lender or project manager cares most about cost and risk. A venture investor cares most about upside and owning a scarce position. I scored both.

Table 6. *Weighted totals under two lenses (out of 5).*

STACK	PROJECT-FINANCE LENS	VENTURE LENS	READING
Alkaline (pressurised)	**3.90**	3.55	Best for the bankable core
Hybrid ALK + PEM	3.80	**3.90**	Best balance; top under the venture lens
PEM	2.85	2.80	Good flexibility, costly and iridium-bound
AEM	1.85	3.00	A real option, not a foundation
SOEC	1.20	1.80	Wrong place, wrong time for us

Project-finance weights: cost 25%, risk 20%, market 15%, value 15%, supply 10%, fit 10%, upside 5%. Venture weights: upside 30%, supply 20%, cost 15%, risk 10%, value 10%, fit 10%, market 5%.

Six lenses on one decision

Macro forces first (STEEPLE, scenarios), then the industry and the firm (Porter, SWOT, VRIO), then what can go wrong (bow-tie). Every lens is scored on the same 1–5 scale.

4.1 STEEPLE: the forces outside our control

PESTEL looks at six outside forces; STEEPLE adds a seventh, ethics. Rather than write two tables that repeat each other, I use STEEPLE, whose first six rows *are* the PESTEL. "Impact" is how strongly the force bears on our decision (5 = very high). "Direction" says whether it helps (+) or hurts (–) a green-hydrogen project today.

Table 7. *STEEPLE scan (Social, Technological, Economic, Environmental, Political, Legal, Ethical).*

FORCE	WHAT IS HAPPENING, WITH EVIDENCE	IMPACT	DIR.	CONF.
Social	The Mission promises over 6 lakh jobs [?]. Farmers are insulated from fertiliser costs by a urea price frozen since 2012 [?], so there is no consumer push-back, but no consumer pull either. Land and water are local flashpoints.	●●●○○	±	MEDIUM
Technological	Electrolyser factories run at about 9% utilisation [?]; alkaline prices halved in three years [?]. AEM and SOEC remain unproven at scale [?, ?].	●●●●○	+	HIGH
Economic	Green ammonia clears about 19% above grey [?]; green hydrogen US$3.5–4 against grey US$2.2 [?]. Gas is about half imported [?] and the rupee has weakened [?], which hurts grey too.	●●●●●	– today, + as hedge	HIGH
Environmental	A legal 2 kg CO_2 ceiling [?]; leakage warms (GWP about 11.6) [?]; water is manageable at 20–30 L/kg [?].	●●●○○	+	HIGH
Political	₹19,744 crore Mission, ₹17,490 crore of it in SIGHT [?]; 862,000 t/yr allocated [?]. Purchase mandates for refineries and fertiliser plants are discussed, not yet notified [?].	●●●●●	+	HIGH
Legal	Green Hydrogen Standard with BEE certification [?]; environmental clearance exemption for green hydrogen and ammonia plants [?]; PESO safety approvals [?]; EU CBAM charging carbon on imported hydrogen and ammonia since 2026 [?].	●●●●○	+	HIGH
Ethical	"Zero emission" claims can mislead: embedded emissions are not counted [?], and Germany's first hydrogen trains ran on a chemical by-product [?]. Public money subsidises private plants; water in drought districts raises fairness questions.	●●●○○	–	MEDIUM

> **WHAT STEEPLE TELLS ME**
>
> The two strongest forces pull in opposite directions: **politics pushes** (5, +) and **economics resists** (5, – today). The project lives in the tension between them. Its survival depends on keeping the political support converted into *signed contracts*, because policy moods change faster than a 25-year plant can. HIGH

4.2 Scenario planning: four futures to 2035

I picked the two uncertainties that matter most and are least predictable. **Across**: how fast the cost of green hydrogen falls (electrolysers, renewables, capital). **Up**: how certain demand is, from voluntary buying to mandated, contracted purchase.

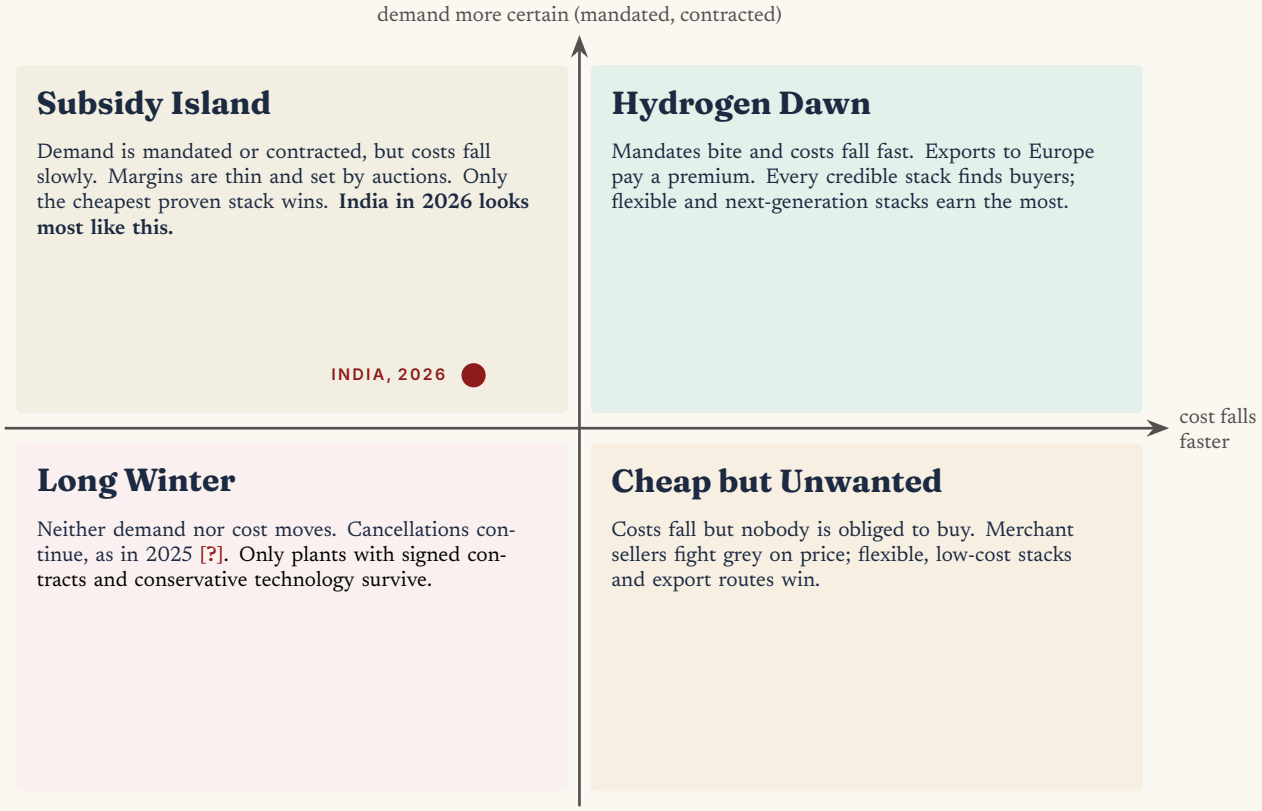


Figure 14. *Four futures for Indian green hydrogen to 2035.*

Table 8. *How well each stack survives each future. Score: 5 = thrives, 1 = stranded.*

STACK	SUBSIDY ISLAND	HYDROGEN DAWN	CHEAP BUT UNWANTED	LONG WINTER	AVERAGE
Alkaline (pressurised)	5	4	3	4	**4.00**
Hybrid ALK + PEM	4	5	4	3	**4.00**
PEM	2	4	4	2	3.00
AEM	1	5	4	1	2.75
SOEC	1	3	2	1	1.75

Alkaline and the hybrid tie on average, but they are robust in different ways. Alkaline is strongest in the futures that look most like today; the hybrid is strongest in the futures we hope for. AEM is the classic option payoff: brilliant in two futures, stranded in two. **Signposts to watch**: notification of green-hydrogen purchase mandates [?]; the next SECI tranche's prices; electrolyser factory utilisation [?]; and the first multi-year field data on AEM durability [?].

4.3 Porter's five forces: is this a good industry to be in?

Here the scores measure the *strength of each force*, so 5 means the force squeezes our profits hard.

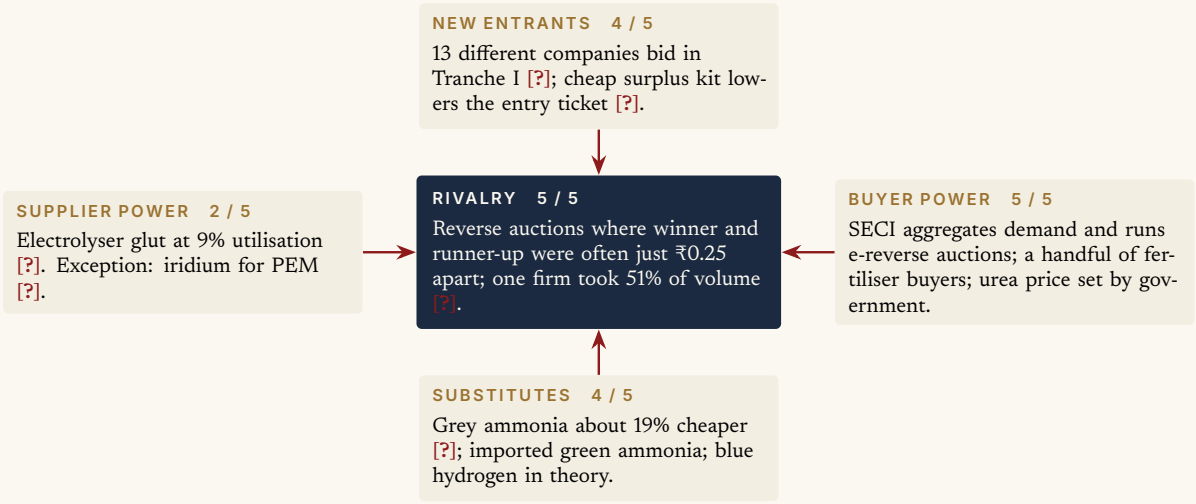


Figure 15. *Five forces acting on a new Indian green-ammonia producer. 5 = the force squeezes margins hardest.*

WHAT PORTER TELLS ME

Four of the five forces score 4 or 5. Structurally, this is a **low-margin, commodity-style industry** in which the buyer sets the price. There are only two ways to earn above-normal returns: be the lowest-cost producer (cheap power, cheap capital, tight capex control), or escape to a less crowded corner, such as 25-year refinery contracts, CBAM-protected exports or certified-green premium buyers. The one force in our favour, weak supplier power, is temporary: it lasts only while the electrolyser glut lasts. HIGH

4.4 SWOT: our venture as a whole

This SWOT is for the venture we are planning: a new Indian producer of green ammonia (and some hydrogen) on a coastal site. Section **??** gives a separate SWOT for each technology bet.

STRENGTHS

- Some of the world's cheapest firmed solar: ₹2.86/kWh with storage [?]
- Contracts already exist: 10-year ammonia, 25-year refinery [?, ?]
- A clear legal definition of "green" [?]
- Coastal sites solve water with cheap desalination [?]

WEAKNESSES

- Replaces only about ₹54 of value per ₹100 spent (my model)
- A new entrant lacks the balance sheet and captive power of large groups
- First-of-a-kind integration of variable power with a steady chemical loop
- 10-year contracts on a 25-year asset leave a 15-year price tail

OPPORTUNITIES

- Refinery demand on 25-year terms at ₹279–397/kg [?, ?]
- EU CBAM makes clean exports pay [?]
- Electrolyser glut means cheaper, better-guaranteed kit [?]
- By-product oxygen and future carbon-market revenue

THREATS

- Winner's curse: rivals bidding below true cost (Section **??**)
- A fall in gas prices widens the grey-green gap
- OEM insolvency in a consolidating industry [?]
- Policy delay; mandates still not notified [?]

4.5 VRIO: what could give us a lasting edge?

VRIO asks four questions of each resource. Is it **V**aluable? **R**are? Costly to **I**mitate? Are we **O**rganised to use it? Only a resource that passes all four gives a lasting advantage. The score is my rating of the advantage it gives us (5 = sustained).

Table 9. *VRIO screen of the resources our venture could build or buy.*

RESOURCE	V	R	I	O	OUTCOME	SCORE
Cheap captive renewables and port land	Y	Partly	Partly	Y	Temporary advantage	●●●●○
A SECI 10-year offtake contract	Y	Y	N	Y	Temporary: it is re-auctioned	●●●○○
The electrolyser stack itself (any type)	Y	N	N	Y	Competitive parity: anyone can buy it	●●○○○
Know-how in running electrolysers flexibly with a steady ammonia loop	Y	Y	Y	To build	**Sustained, if we organise for it**	●●●●●
Cheap long-tenor capital	Y	Y	Y	Depends on sponsor	Advantage for big groups, a gap for us	●●○○○
An hourly, auditable green-certification data trail (for CBAM and buyers)	Y	Y	Partly	To build	Temporary, possibly sustained	●●●○○
An AEM technology licence or partnership	Uncertain	Y	Y	N	Unrealised option	●●○○○

KEY FINDING • THE MOAT IS NOT THE MACHINE

Electrolysers are becoming a commodity; anyone can buy one. What is rare and hard to copy is **the skill of running a variable-power hydrogen plant against a steady chemical process without losing efficiency or breaking equipment**, plus a clean data trail that proves the product is green hour by hour. That points to an uncomfortable conclusion for a technology bet: *our edge will come from operations and data more than from the stack we pick.*

MEDIUM

4.6 Bow-tie: what can go wrong, and what stops it

A bow-tie puts one feared event in the middle. Threats that could cause it sit on the left, each with a preventive barrier. Consequences if it happens sit on the right, each with a recovery barrier. I chose the event that would kill the project financially, not the most dramatic one.

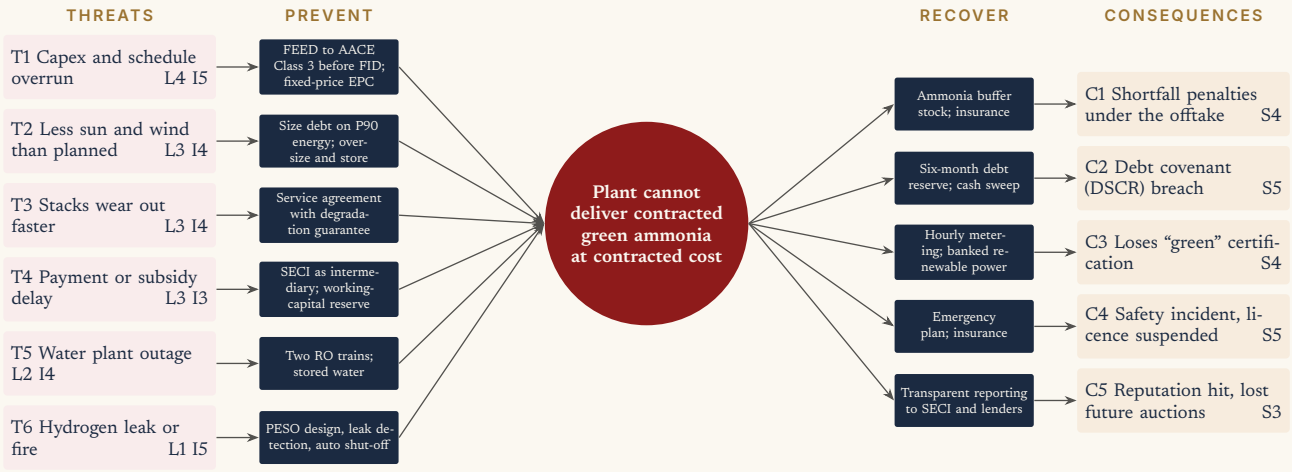


Figure 16. *Bow-tie for the project's critical event. L = likelihood, I = impact, S = severity, each on 1–5 (5 = worst).*

Table 10. *Threats ranked by risk score (likelihood × impact, maximum 25).*

THREAT	L	I	SCORE	WHY I RATE IT SO	CONF.
T1 Capex and schedule overrun	4	5	**20**	Capex is the largest cost swing (US$1.30/kg); an early estimate can run 30–100% high [?]	MEDIUM
T2 Renewable shortfall	3	4	12	Electricity is the second-largest swing (US$0.84/kg)	MEDIUM
T3 Faster stack wear	3	4	12	No long field record at variable load for newer stacks [?]	MEDIUM
T4 Payment or subsidy delay	3	3	9	State intermediary reduces but does not remove it	LOW
T5 Water outage	2	4	8	Small cost, but it stops production	LOW
T6 Leak or fire	1	5	5	Low frequency, catastrophic severity; well regulated [?]	MEDIUM

The bow-tie confirms what the tornado chart said: **the most dangerous threat is a project-management one.** An AACE Class 5 (concept-stage) estimate can be off by −20% to −50% on the low side and +30% to +100% on the high side; a Class 3 estimate, typically used to authorise the budget, narrows that to roughly −10/−20% and +10/+30% [?]. Getting from one to the other before the investment decision is the cheapest risk reduction available to us. HIGH

THE BET

What each technology offers, and where I lean

A SWOT for every stack, the options on the table, my recommendation, and how sure I am of each finding.

5.1 What each bet offers

Table 11. *A SWOT for each technology bet, compressed to the point that matters most in each box.*

BET	STRENGTH	WEAKNESS	OPPORTUNITY	THREAT
Alkaline	Cheapest kilogram (US$3.83 base, P50 US$4.07); most bankable; Indian factories coming	Wastes the shoulders of the solar day; high minimum load	Glut gives buying power on price and warranty	Chinese concentration and OEM failures; rupee risk
Hybrid ALK + PEM	Near-alkaline cost (+US$0.19/kg) with better use of renewable hours; top under the venture lens	Two OEMs, harder integration, one more failure point	The PEM slot can later host AEM modules: a built-in switching option	Integration risk lands on us unless one EPC wraps it
PEM	Best at following sun and wind; compact; mature	US$1.26/kg dearer than alkaline; iridium bound	Iridium thrifting could cut cost sharply	Iridium supply ceiling of a few GW a year
AEM	No precious metals; PEM-like flexibility; an Indian factory is planned	Unproven durability; small makers; US$2.00/kg dearer today	If it wins, a scarce IP and manufacturing position	Stranded in two of four futures; makers may not survive
SOEC	Highest electrical efficiency	Needs constant heat; short life; leading maker paused	Pairing with industrial waste heat in the 2030s	Market not ready; wrong fit for solar

One result changed how I think about the AEM bet. I asked what happens if AEM delivers everything its backers promise by the early 2030s: US$700/kW installed, 60,000-hour stacks, 52 kWh/kg. It would make hydrogen at about **US$2.97/kg**. But alkaline under the *same* favourable conditions (US$800/kW, cheaper power, slightly cheaper capital) would make it at about **US$3.03/kg**. Even a winning AEM roughly ties a maturing alkaline on cost. MEDIUM

> KEY FINDING • AEM IS A SOVEREIGNTY BET, NOT A COST BET
>
> The case for AEM is not that it will make cheaper hydrogen; it probably will not be much cheaper. The case is that it needs *no iridium, no platinum and no Chinese supply chain*, and that whoever masters it owns scarce intellectual property and manufacturing know-how. That is a venture-capital thesis, and it should be priced like one: small, staged and optional. It should not carry a bank-financed 100 MW plant. MEDIUM

5.2 The options on the table

Table 12. *The five realistic options for our plant.*

OPTION	CHOOSE IT IF…	WHAT YOU GIVE UP	PF / VC SCORE
A. Bank on alkaline 100 MW pressurised alkaline	The priority is the lowest cost and the easiest loan. Best fit for "Subsidy Island"	Some renewable hours; upside from new technology	3.90 / 3.55
B. Build the hybrid 80 MW alkaline + 20 MW PEM	You want near-lowest cost *and* flexibility, plus a slot for future technology	About US$0.19/kg and some integration simplicity	3.80 / 3.90
C. All PEM	Iridium-light stacks arrive below about US$1,300/kW with strong warranties	About US$1.26/kg today	2.85 / 2.80
D. AEM as a call option a 2–5 MW pilot bay beside the main plant	You want exposure to the next technology without betting the plant	A small, ring-fenced sum	1.85 / 3.00
E. SOEC on watch	Only where steady industrial waste heat is available	n/a for us now	1.20 / 1.80

WHERE I LEAN, BEFORE WE DECIDE TOGETHER

Bank on alkaline, build for the hybrid, hold AEM as a call option. In practice that is Option B with Option D attached: 80 MW of pressurised alkaline as the bankable core, 20 MW of PEM to soak up the variable hours, and a small, separately funded AEM pilot bay. It costs about 5% more per kilogram than pure alkaline in my base case, and in return it buys three things a lender and an investor both value: better use of the power we pay for, resilience if one OEM fails, and a ready-made slot to switch technology later without rebuilding the plant.

I would set clear triggers for exercising the AEM option, not hopes: *at least 20,000 hours of field data on a multi-megawatt unit, installed cost below about US$800/kW, and a supplier with a balance sheet that can stand behind a warranty.* If those triggers are not met by the first stack replacement (around year eight to ten), the bay is repurposed as more PEM and the option simply expires. That is how a finance person turns a technology dream into a decision with a known maximum loss. MEDIUM

Option A is a perfectly defensible choice, and if the project brief puts the weight on bankability and simplicity, I would not argue hard against it. The decision is ours to make as a team, and the rest of the project (schedule, risk register, financial model, real-option valuation) will be built on whichever stack we choose.

5.3 Key findings

1. **Hydrogen is a chemical first and a fuel second.** India already uses about 6.7 Mt a year, almost all of it grey; the quickest climate win is cleaning that up, not new hydrogen gadgets [?].
2. **Hydrogen trains and cars are symbols, not the main market.** Indian rail is 99.2% electrified [?], and battery cars use electricity two to three times as efficiently [?].
3. **India pays for grey hydrogen twice**: about 55–70 Mt of CO_2 a year (my estimate), and a fertiliser subsidy of ₹1.86 lakh crore exposed to imported gas and a weak rupee [?, ?].
4. **Green hydrogen is a hedge**: it turns a volatile, dollar-linked cost into a fixed rupee cost.
5. **Alkaline makes the cheapest kilogram**: about US$3.83 in my base case and US$4.07 at the median, matching the market's US$3.5–4 [?].
6. **The auctions assume about US$2.5–2.9/kg**, which needs almost every assumption to go right: a winner's-curse risk worth watching [?].

7 **Capex is the biggest lever** (US$1.30/kg swing), so cost control is a project-management problem before it is a technology problem.

8 **No stack pays for itself yet**: the best replaces about ₹54 of grey hydrogen per ₹100 spent. The offtake contract is the business.

9 **The industry structure is harsh**: four of Porter's five forces score 4 or 5. Advantage must come from cost position and operating know-how, not from the machine.

10 **AEM is a sovereignty bet, not a cost bet**: worth owning as a small option, not as the foundation.

5.4 How sure am I? A confidence ledger

Different studies give different numbers, so here is where I would and would not stake the project.

Table 13. *Confidence in the dossier's main claims.*

CLAIM	CONFIDENCE	WHY
Grey hydrogen emits 10–12 kg CO_2 per kg; world production released 920 Mt in 2023	HIGH	IEA primary data [?]
SECI Tranche I prices, winners and volumes	HIGH	Read directly from SECI's own result sheet [?]
Mission targets and outlay	HIGH	Cabinet approval figures [?]
Battery cars are 2–3 times more efficient than fuel-cell cars	HIGH	Consistent across sources; VW figures [?]
India's current hydrogen use is about 6–7 Mt	MEDIUM	Modelled from chemistry in 2021, not measured [?]
Alkaline LCOH of about US$3.8–4.1/kg	MEDIUM	My model; calibrates to analyst figures [?], but inputs are ranges
Auctions imply US$2.5–2.9/kg hydrogen	MEDIUM	Depends on my ammonia-loop cost assumption
Stack rankings (alkaline > hybrid > PEM > AEM > SOEC on cost)	HIGH	Robust across all 100,000 simulations
Electrolyser factory glut (about 9% utilisation)	HIGH	IEA via Argus [?]
AEM commercial status and Indian factory plans	LOW	Industry aggregator; few primary sources [?]
Cost of capital 8.5–12.5%; renewable hours 55–68%	LOW	My assumptions; need lender and site data
Long-lead times for compressors and switchgear	LOW	From the Gemini brief, not independently verified [?]

What I checked, and how the model works

Annex A corrects the Gemini brief where it was wrong; Annex B lists every model input; Annex C says how to rerun it.

Annex A • Fact-check of the Gemini research brief

The Gemini brief [?] was a useful map but not a source. I checked each claim I reused against primary or close-to-primary sources. Most held up. These did not, or needed qualifying.

Table 14. *Gemini claims, verdicts and corrections.*

WHAT GEMINI SAID	VERDICT	WHAT IS ACTUALLY TRUE
SIGHT subsidy is ₹8.82 per kg *of hydrogen*, falling to ₹5.30	**Wrong unit**	It is per kg *of green ammonia*: ₹8.82, ₹7.06, ₹5.30 over three years [?]. Since a kilogram of hydrogen makes about 5.6 kg of ammonia, the error overstates the subsidy about 5.6 times.
Nine Mode-2A auction results	**Incomplete**	There were 13 auctions totalling 724,000 t/yr. Missing: GNFC Bharuch (Onix, ₹52.50), Coromandel Vizag (ACME, ₹51.89), MBAPL-III Dhule (SCC, ₹53.05), Paradeep Phosphates Zuarinagar (ACME, ₹62.84). The nine listed prices are correct [?].
Exchange rate of ₹83.50 per dollar	**Outdated**	The rupee traded around ₹86.6–91.9 in 2025–26 [?]. I convert auction prices at ₹87 and plan at ₹90.
"Over 400,000 t" of production capacity allocated (its Section 4.4)	**Inconsistent**	Contradicts its own earlier figure. MNRE: 862,000 t/yr allocated to 19 companies [?].
Chinese alkaline at US$300–500/kW "installed"	**Mislabelled**	That is roughly an equipment price. IEA installed cost is US$500–1,100/kW in China and US$1,900–2,500/kW for systems made outside China [?].
Model grey ammonia as a "mean-reverting geometric Brownian motion"	**Contradiction**	Geometric Brownian motion does not mean-revert. Commodity prices are better modelled with an Ornstein–Uhlenbeck process on the log price (the Schwartz one-factor model). I flag this for the GitHub model.
FID needs an AACE Class 2 or Class 1 estimate	**Overstated**	AACE places budget authorisation typically at Class 3 (10–40% definition); Class 2 is for control and bids [?].
Megaprojects overrun 33%, slip 30%, 64% suffer production shortfalls	**Unverified**	I could not open the cited paper. The figures are plausible for oil-and-gas megaprojects but untested for hydrogen. Not used in my analysis. LOW
SECI contracts are 10 years, creating a 15-year merchant tail	**True, but partial**	True for ammonia [?]. It misses that refinery contracts run 25 years [?, ?], which changes the merchant-risk picture for a hydrogen seller.

WHAT GEMINI SAID	VERDICT	WHAT IS ACTUALLY TRUE
PEM can be built at only about 3–7.5 GW a year because of iridium	**Consistent**	Matches 7–8 t/yr of iridium at about 400 kg/GW, assuming PEM gets 20–40% of supply [?].
Green Hydrogen Standard of 2 kg CO_2e/kg	**Verified**	Well-to-gate, 12-month average, BEE as nodal authority [?].
Grey ammonia at US$515/t; green premium 10–24%	**Verified**	US$515/t [?]; lowest bid 10% above grey [?]; my volume-weighted figure is 19%.
Hydrogen trains, cars and daily-life uses	**Absent**	Not covered. Added in Phase 1.

Annex B • Model inputs

Every uncertain input is drawn from a triangular distribution (low, most likely, high). I chose triangular rather than a fancier shape because the evidence gives ranges, not shapes; claiming to know the tails better than the sources do would be false precision.

Table 15. *Inputs shared by all stacks.*

INPUT	LOW	LIKELY	HIGH	BASIS
Electricity, hybrid renewables (US$/MWh)	32	38	48	Solar + storage at ₹2.86/kWh [?]; firm power at ₹5.25 [?]
Cost of capital (WACC)	8.5%	10%	12.5%	My assumption; Gemini estimated 9.5–11.5% [?]
Share of hours renewables can feed the stack	55%	62%	68%	My assumption for an oversized solar + wind hybrid
Plant life		25 years		Standard infrastructure life
Water		US$0.03/kg		Purification at most US$0.10/kg [?]
Exchange rate		₹90 per US$		Rupee 86.6–91.9 in 2025–26 [?]
Grey hydrogen benchmark		US$2.1–2.2/kg		[?, ?]

Table 16. *Stack-specific inputs (low / likely / high).*

STACK	INSTALLED US$/KW	KWH PER KG	STACK LIFE (H)	REPL. % CAPEX	O&M %/YR	HOURS USED
Alkaline	800 / 1,100 / 1,600	50 / 53 / 57	60k / 75k / 90k	35	2.0	92%
Hybrid	900 / 1,250 / 1,750	50.5 / 53.2 / 57.2	55k / 72k / 88k	35	2.1	97.5%
PEM	1,300 / 1,800 / 2,400	51 / 54 / 58	40k / 60k / 80k	35	2.5	99%
AEM	1,000 / 1,700 / 3,000	51 / 55 / 62	15k / 30k / 50k	40	3.0	98%
SOEC*	2,800 / 3,800 / 5,600	37 / 40 / 44	10k / 20k / 30k	45	4.0	80–90% CF

**SOEC runs on firm power at US$50/58/66 per MWh plus 8 kWh of heat per kg at US$12/MWh. Capex ranges anchor on IEA 2025 installed costs [?], IEA's earlier technology ranges [?] and IRENA [?]; efficiency and life on IRENA [?]. "Hours used" is the share of renewable-available hours each stack can capture, reflecting minimum-load limits [?].*

Annex C • Rerunning the numbers

The analysis lives in three short Python files, ready for our GitHub repository: `model.py` (levelised cost, Monte Carlo with 100,000 draws per stack, tornado, SECI back-calculation), `believe.py` (the "what you

have to believe" ladder and the AEM comparison) and `charts.py` (every chart in this dossier). The random seed is fixed at 20261013, so anyone running the files gets exactly the numbers printed here. The obvious next upgrades are a proper project-finance cash-flow waterfall (debt service cover, loan-life cover), a mean-reverting model for gas and ammonia prices, and hourly solar and wind profiles for the four candidate ports instead of a single "hours available" number.

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APA style. Every bracketed number in the text jumps here; every entry links to the original page. Accessed October 2026.

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